



Build Bit's Stack

Quick facilitation notes & answer key

Goal

Children compose their own short block program — choosing and ordering move blocks — to make Bit reach the star, then run it to check. This is the writing side of coding: turning a goal ('get to box 4') into an ordered set of instructions and testing it.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children plan and create a sequence of steps to reach a goal, then test and refine it.

No reading required of the child: read each prompt aloud; the child draws an arrow (➡ or ⬅) in each empty block.

Materials

The printed worksheet and a pencil. Optional: floor boxes 1-5 with a movable star so children can walk and test the program they built.

Run it (about 15 min)

1. MODEL: for Build 1, count the boxes from Bit (1) to the star (4): that is 3 steps right, so draw ➡ ➡ ➡. Run it together to confirm box 4.
2. GUIDED: Exercise 1 — children fill the 3 blocks with ➡. Exercise 2 — the star is to the LEFT, so the blocks are ⬅ ⬅.
3. YOU TRY: Exercise 3 — box 1 to box 3 is 2 steps right: ➡ ➡.
4. Connect: 'You wrote the program — then you ran it to check. That is exactly what coders do.'

Answer key (modelled by the run-gate)

Ex 1 — Build 1: ➡ ➡ ➡ (box 1 → 2 → 3 → 4, the star).

Ex 2 — Build 2: ⬅ ⬅ (box 4 → 3 → 2, the star).

Ex 3 — Build 3: ➡ ➡ (box 1 → 2 → 3, the star).

Watch for & differentiate

Most common slip: too few or too many arrows — have the child count boxes from Bit to the star first, then draw that many.

Direction check: arrows point the way Bit travels; toward the star on the right is ➡, on the left is ⬅.

Simplify: do Builds 1 and 3 only (both go right). Pre-count the boxes together.

Extend: ask for a different program that also reaches the star (e.g. ➡ ➡ ⬅ ➡ for Build 3) and run it to test — there can be more than one.

Movement option: the child calls out their arrows while a partner walks the boxes to test the program.

Success looks like

The child fills every empty block with an arrow, lands Bit on the star for each build, and can run the stack to prove it works.